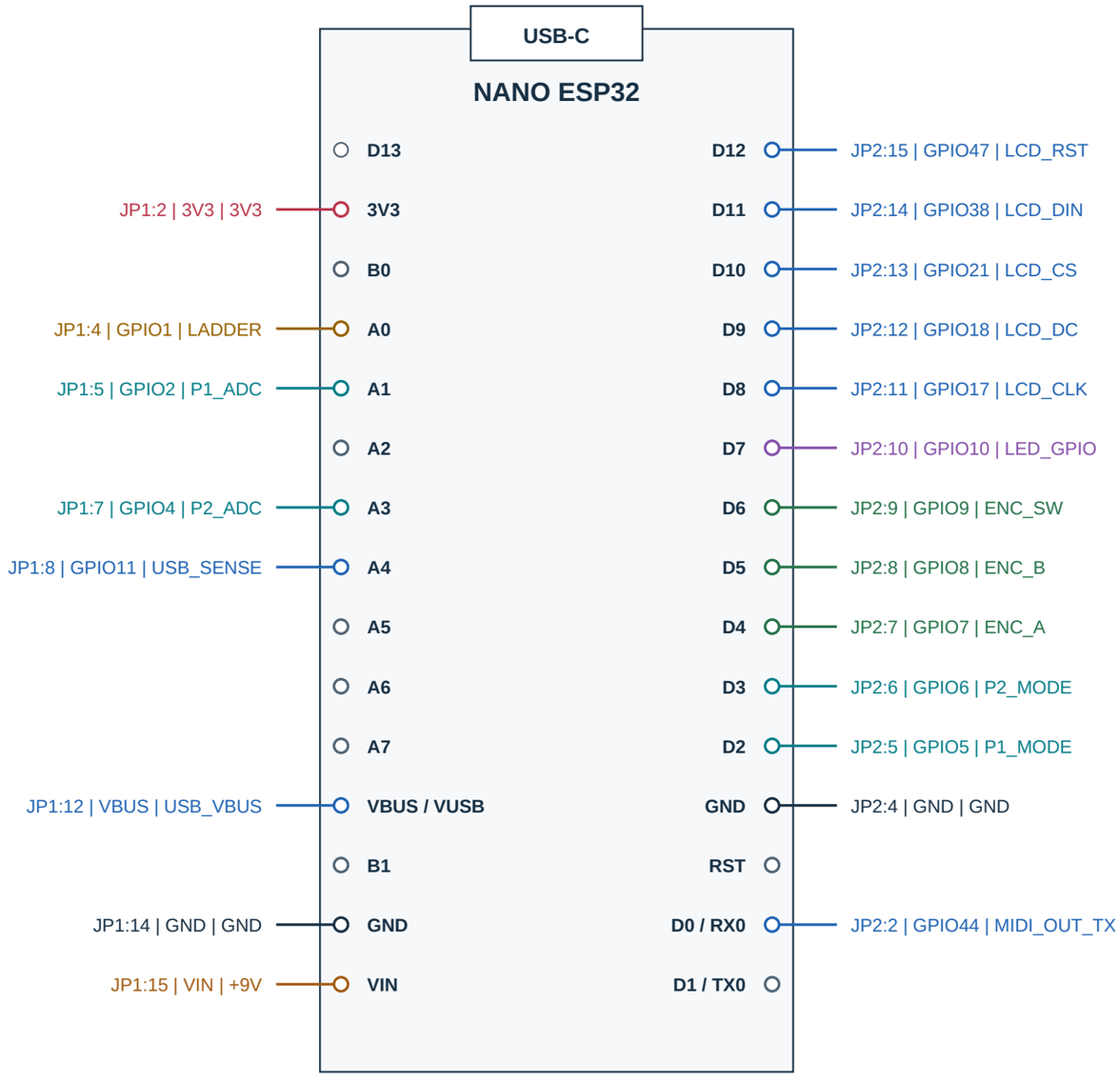
# Controller and display - external pins

Arduino Nano ESP32 / ABX00083. Purchased boards stop at external pins; logical GPIO names accompany used board labels.

## U1 Arduino Nano ESP32

Component-side top view, USB-C up. All printed labels retained; unused outside callouts omitted.




JP1: 1-15 top-down; JP2: 15-1 top-down (official schematic pad numbering).

VBUS is labelled VUSB in the official pinout. RST is RESET in that pinout.

Native USB MIDI uses the USB-C connector. B1 / GND + RST enters the ROM flasher.

## U2 Waveshare 24382

P2 - 8-pin GH1.25 connector

| Pin | LCD label | Nano label / GPIO |
|---|---|---|
| 1 | BL | 3V3 |
| 2 | RST | D12 / GPIO47 |
| 3 | D/C | D9 / GPIO18 |
| 4 | CS | D10 / GPIO21 |
| 5 | SCL / CLK | D8 / GPIO17 |
| 6 | SDA / DIN | D11 / GPIO38 |
| 7 | GND | GND |
| 8 | VCC | 3V3 |

SCL and SDA are SPI signals here.

No MISO connection is required.

BL ties to VCC (3V3) for always-on backlight.

Use the external P2, not the LCD FPC.

### Display orientation

Landscape 280 x 240 pixels.

Left variant: firmware rotation 180 degrees.

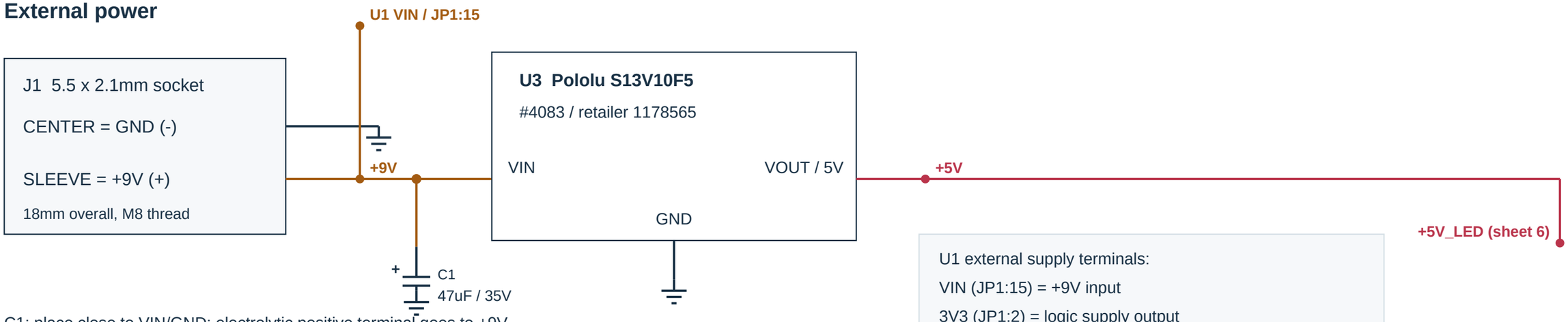
Right variant: firmware rotation 0 degrees.

The electrical wiring stays identical.

# Power distribution and rotary encoder

9V center-negative input; 3V3 logic; VIN receives 9V; converter 5V powers only optional LEDs. USB VBUS is sensed, not joined to 5V.

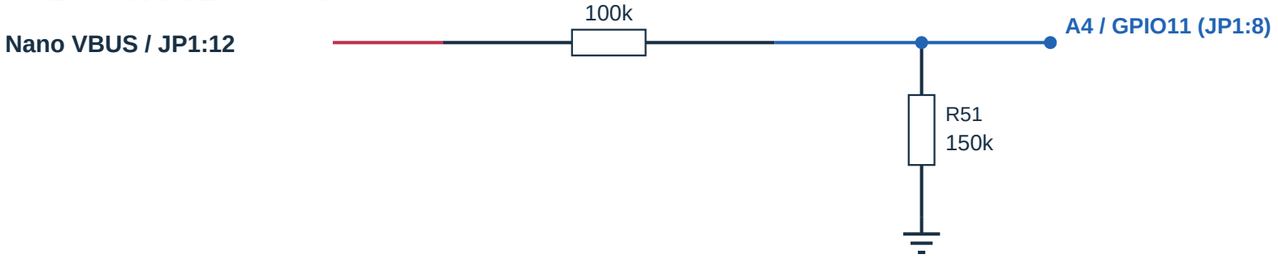
## External power




C1: place close to VIN/GND; electrolytic positive terminal goes to +9V.

U1 external supply terminals:
VIN (JP1:15) = +9V input
3V3 (JP1:2) = logic supply output
GND (JP1:14 / JP2:4) = common return

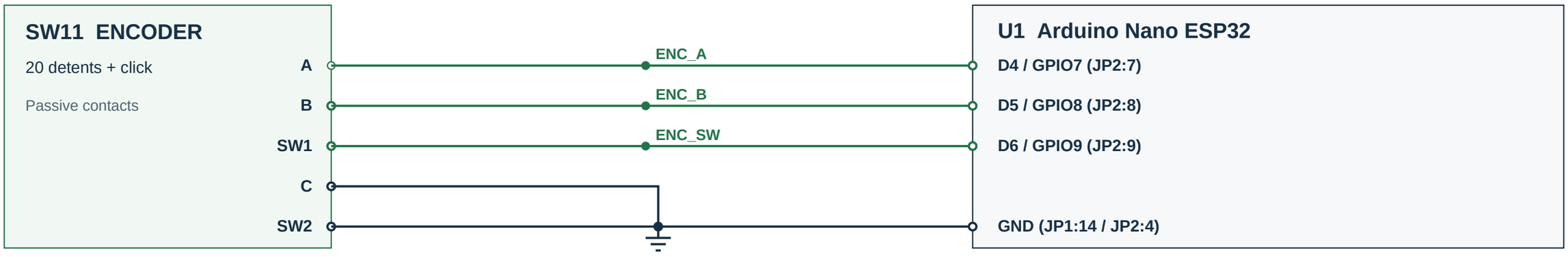
## USB-C and VBUS sense




Native USB-C: MIDI / ROM serial flashing.
B1 to GND, tap RST, remove jumper to flash.
VBUS divider: 3.0V at 5.0V USB input.
Keep converter +5V separate from USB VBUS.
USB-only power leaves the LED rail off.

## SW11 20-detent encoder with click - direct wiring

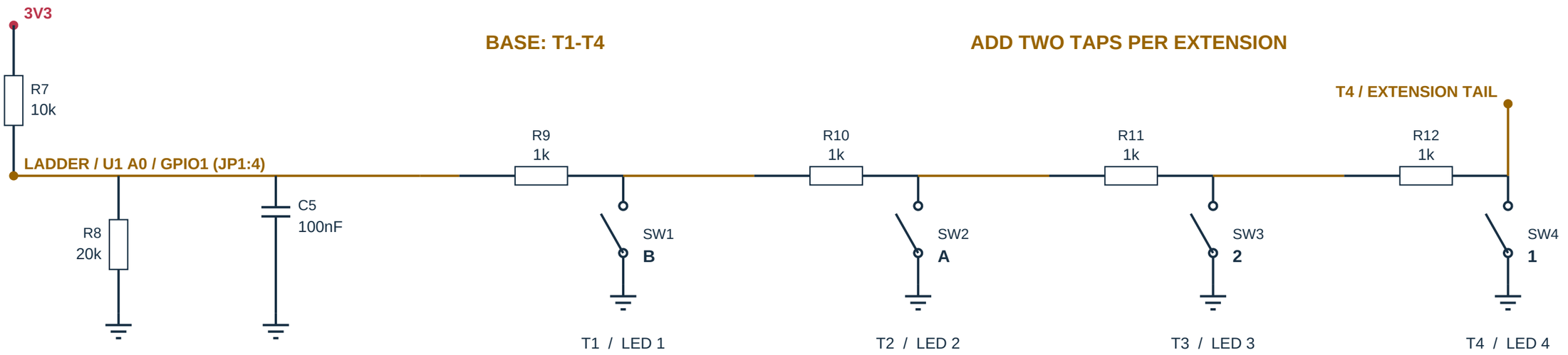
No external resistors or capacitors. Firmware enables internal pull-ups on GPIO7, GPIO8 and GPIO9.

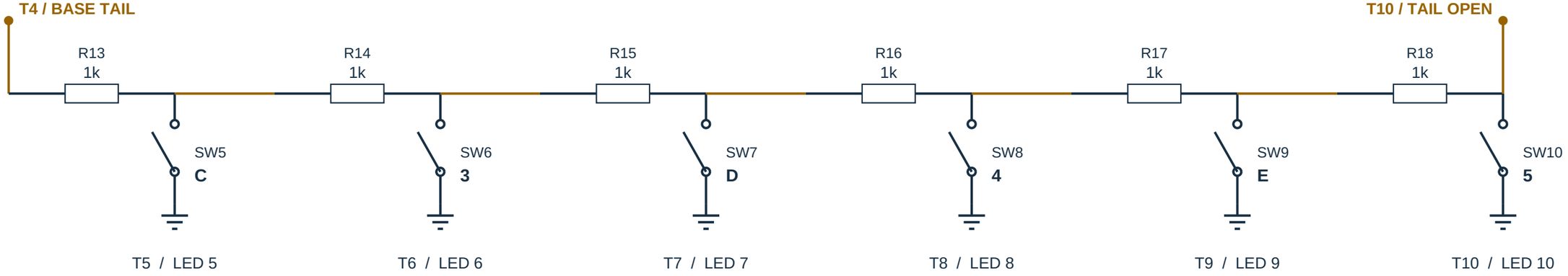



Functional pin arrangement: verify A/B/C and the two click terminals on the selected encoder.
Software decodes A/B and debounces the click. Keep the encoder harness short.

# Modular footswitch resistor ladder

A single series-tap ladder supports 4, 6, 8 or 10 normally-open switches. Only one switch is detected at a time.




## Modular harness and numbering

Each extension receives TAIL + GND and adds 2 x 1k.

Optional LED harness adds +5V and DOUT -> next DIN.

No terminating resistor at the final tail. All resistors 1%.

Select installed count in software, then capture calibration.

Left/right enclosure setting changes visual order only.

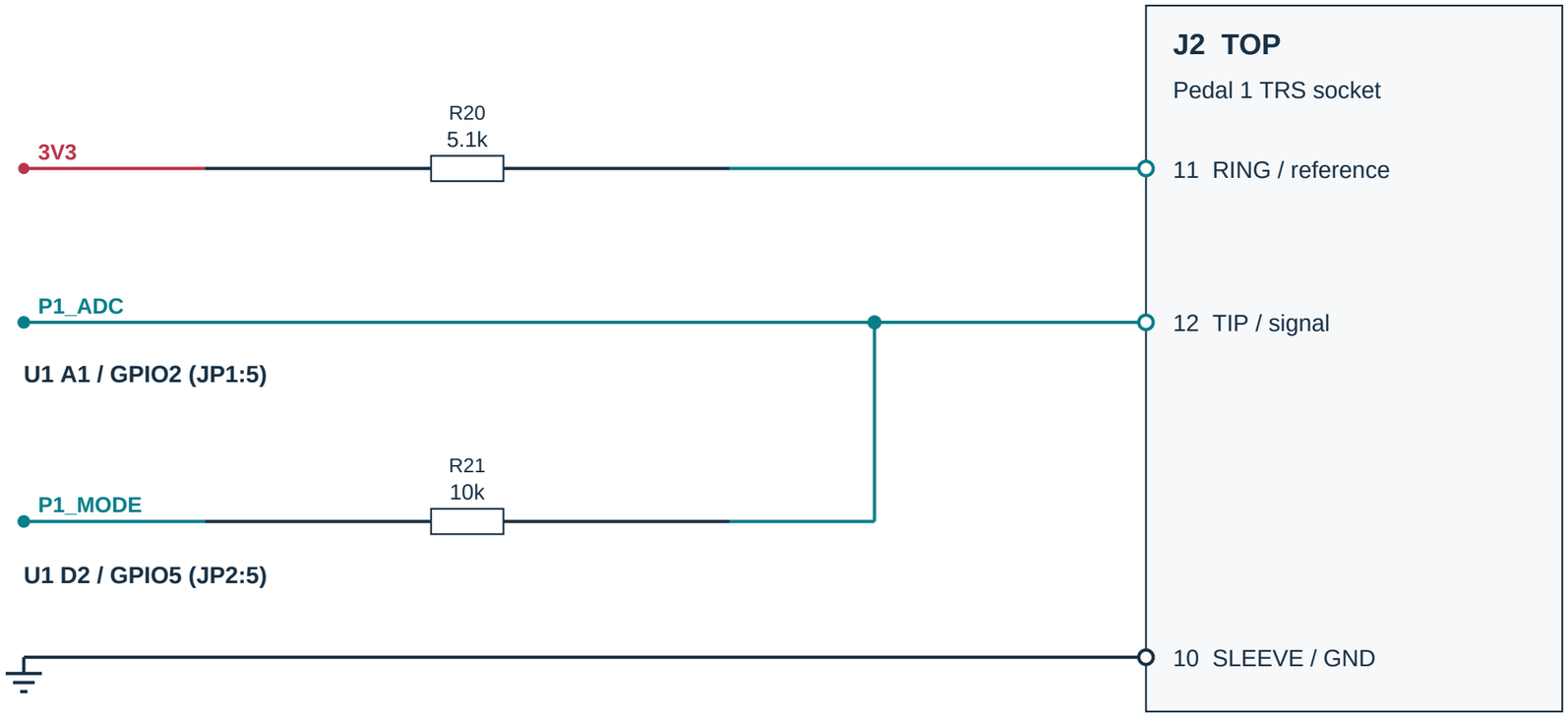
| Tap | Name / LED | Expected mV | Tap | Name / LED | Expected mV |
|---|---|---|---|---|---|
| 1 | B / 1 | 287 | 6 | 3 / 6 | 1042 |
| 2 | A / 2 | 508 | 7 | D / 7 | 1127 |
| 3 | 2 / 3 | 683 | 8 | 4 / 8 | 1200 |
| 4 | 1 / 4 | 825 | 9 | E / 9 | 1264 |
| 5 | C / 5 | 943 | 10 | 5 / 10 | 1320 |

Idle: 2200mV at 3.300V. Values are nominal; calibrate the assembled ladder.

# Continuous pedal input 1 - passive

Two resistors inside the controller. External pedal: passive variable resistor or potentiometer, 0-10k.

## Inside the controller - only two resistors




### Firmware bias state

Resistor mode: mode GPIO = HIGH.

Pot mode: mode GPIO = high impedance.

Disabled / boot: high impedance.

Internal pull-up and pull-down OFF.

Never drive the bias GPIO LOW.

### Nominal values at 3.300V

10k variable resistor: TIP = 1650mV.

10k pot, full travel: TIP = 2185mV.

Ring short current: about 0.65mA.

ADC measures TIP directly (gain = 1).

### Option A - three-wire potentiometer pedal




Capture actual heel and toe voltages after connecting.

### Option B - two-wire variable resistor pedal




RING unused; a TS plug may connect it to SLEEVE.

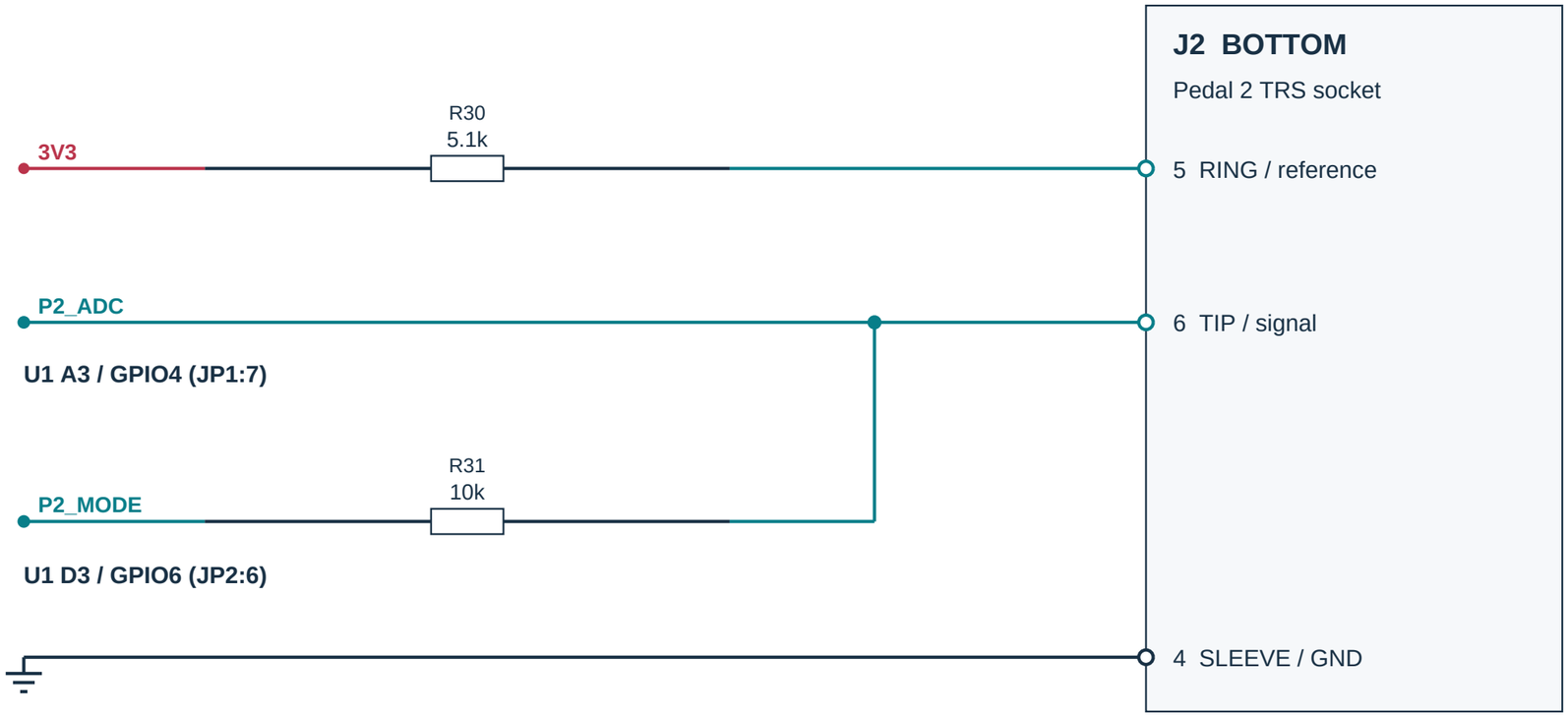
Select resistor mode and configure maximum resistance.

No active pedal interface, scaling divider or external filter capacitors. J2 switched contacts remain NC; pad13 mechanical.

# Continuous pedal input 2 - passive

Two resistors inside the controller. External pedal: passive variable resistor or potentiometer, 0-10k.

## Inside the controller - only two resistors




### Firmware bias state

Resistor mode: mode GPIO = HIGH.

Pot mode: mode GPIO = high impedance.

Disabled / boot: high impedance.

Internal pull-up and pull-down OFF.

Never drive the bias GPIO LOW.

### Nominal values at 3.300V

10k variable resistor: TIP = 1650mV.

10k pot, full travel: TIP = 2185mV.

Ring short current: about 0.65mA.

ADC measures TIP directly (gain = 1).

### Option A - three-wire potentiometer pedal




Capture actual heel and toe voltages after connecting.

### Option B - two-wire variable resistor pedal




RING unused; a TS plug may connect it to SLEEVE.

Select resistor mode and configure maximum resistance.

No active pedal interface, scaling divider or external filter capacitors. J2 switched contacts remain NC; pad13 mechanical.

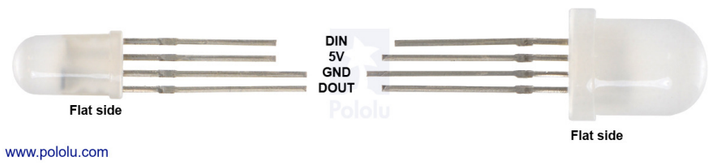
# Pololu 2535 - direct LED chain

Only LEDs and wires: Nano D7 / GPIO10 drives the first DIN directly. All boxes use the manufacturer side-view lead order.

**U1 Arduino Nano ESP32**

- D7 / GPIO10 (JP2:10)
- GND (JP1:14 / JP2:4)

## Pololu package orientation




Manufacturer side view: lens left, flat side down.

## Direct wiring

D7 / GPIO10 -> LED1 DIN.

Each DOUT -> next LED DIN.

Every 5V lead -> regulated +5V.

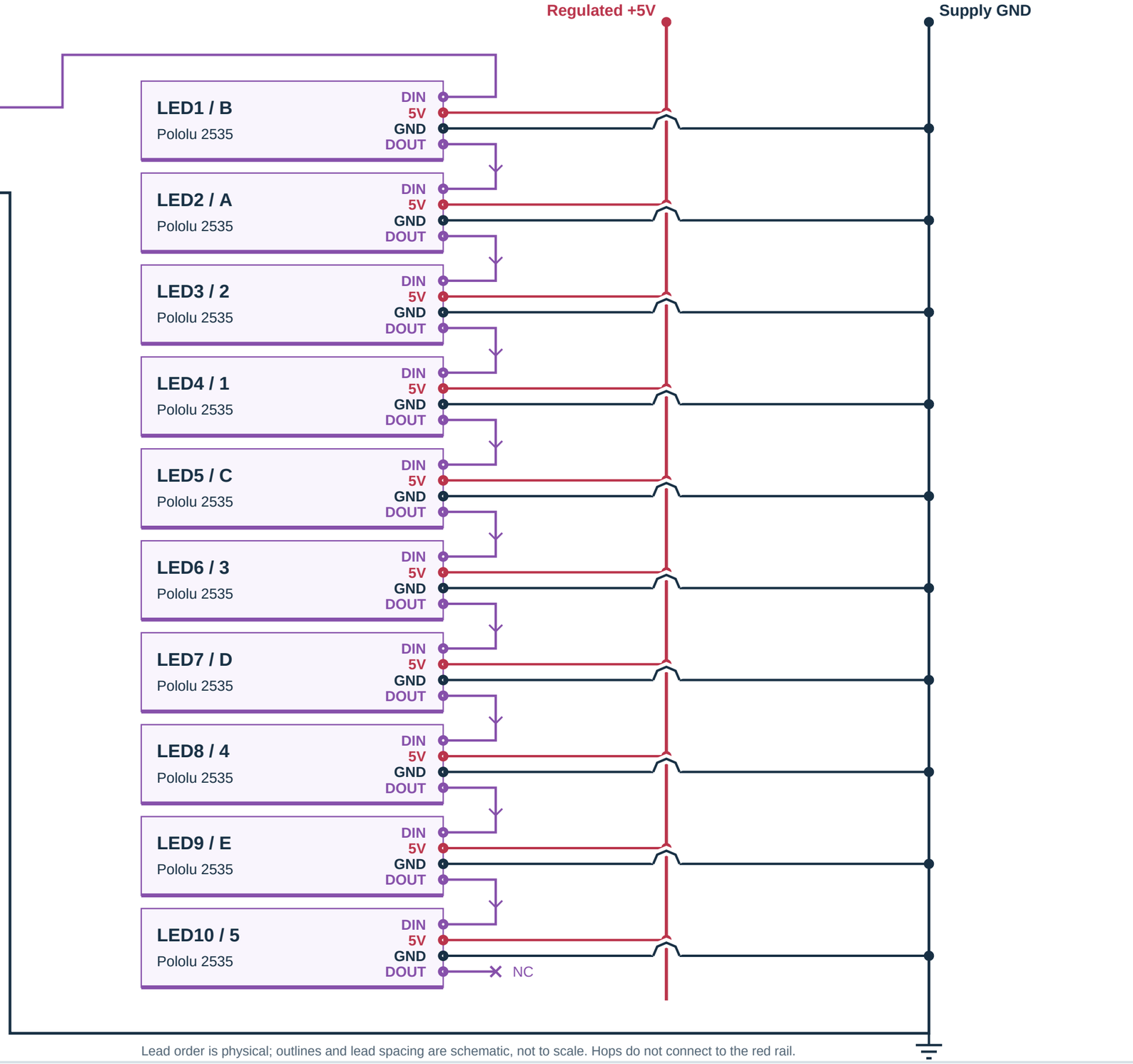
Every GND lead -> common GND.

Last DOUT -> leave unconnected.

RGB / 2535 / 800 kbit/s.

The driver is integrated into each LED.

Direct 3.3V GPIO drive at a 5V LED supply
requires verification with the actual LED batch.
No additional LED-chain components are fitted.

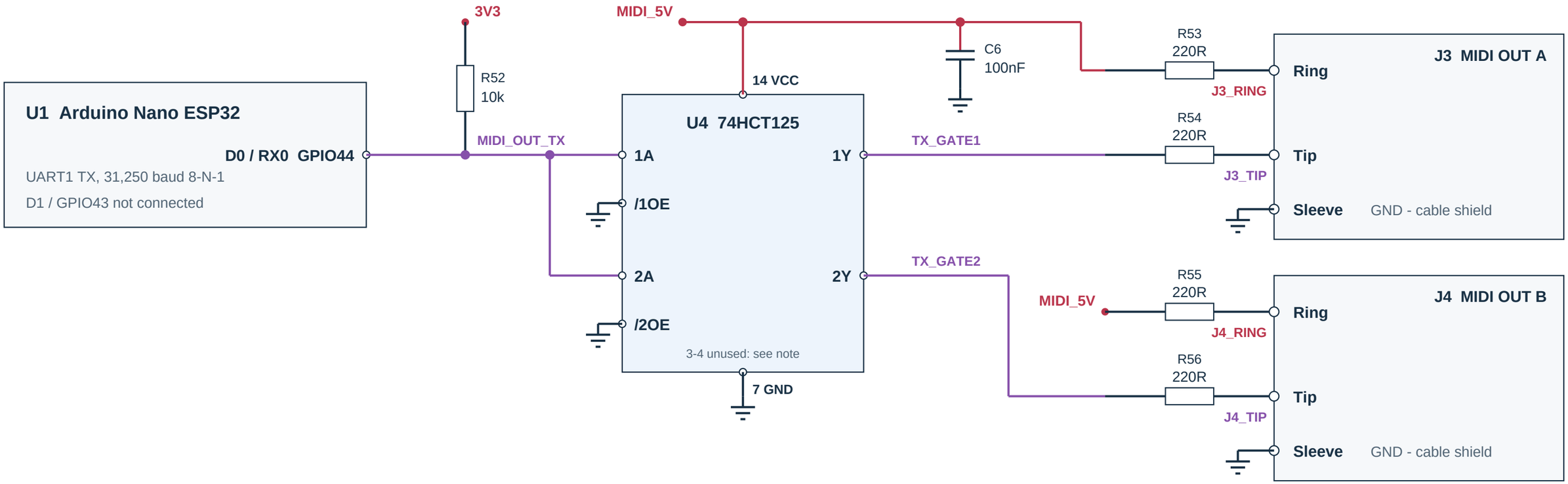



Lead order is physical; outlines and lead spacing are schematic, not to scale. Hops do not connect to the red rail.

# MIDI OUT - two 3.5 mm TRS current loops

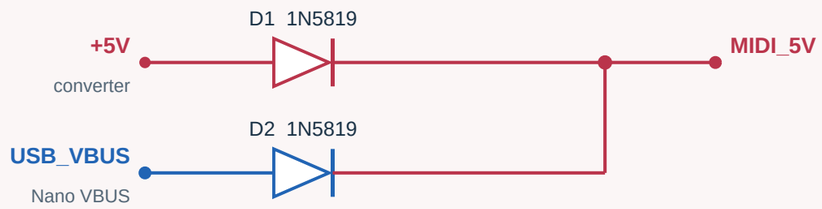
UART1 TX at 31,250 baud, 8-N-1; 74HCT125 gates 1 and 2 each drive one TRS Type A output. UART0 remains ROM-flash serial.

## UART TX, idle pull-up and 74HCT125 buffer




## J3 / J4 3.5 mm TRS, Type A

## MIDI_5V supply / Schottky diode OR




## Buffer and supply notes

R52 holds both buffer inputs idle-high during reset.
U4 /1OE and /2OE grounded; C6 directly across pins 14 / 7.
Unused inputs 9 / 12 and /OE 10 / 13 to GND;
outputs 8 / 11 left open.
Diodes isolate converter +5V and USB VBUS.

Each loop: MIDI_5V -> 220R -> ring -> receiver optocoupler -> tip -> 220R -> buffer output sink (J3: R53/R54/1Y, J4: R55/R56/2Y).

TRS Type A: tip = DIN pin 5, ring = DIN pin 4, sleeve = DIN pin 2. Each sleeve connects to GND to ground the cable shield.

One-way outputs only; no receive connection to the Nano. Use Type A adapters or cables for 5-pin DIN receivers.

# Assembly reference, options and sources

The accompanying netlist and connection schedule include every external component terminal shown in this set.

## Harness and assembly rules

All GND symbols are the same electrical net. Run analog returns separately to the controller ground point.

Keep LADDER and pedal signals away from LED power/data and the DC converter. Pair signals with ground.

Feed LED +5V/GND directly from the power distribution, rather than through the Nano headers.

The 10-pixel chain can draw about 0.5A at full white. Keep total load within the 1A converter rating.

LED lead order, lens pointing left and leads right, flat side down: top DIN, +5V, GND, bottom DOUT.

The LED chain contains only Pololu 2535 LEDs and wiring: GPIO10 directly to the first DIN.

Encoder terminal labels are functions; a specific encoder package pin numbering has not been selected.

GPIO numbers are logical; D/A labels and SCK/MOSI/TX names identify the external board pins.

## Prototype verification

With power disconnected: continuity-check TRS terminals, power polarity, ground and all off-sheet nets.

Verify regulated +5V, then U1 3V3. Flash the passive-input firmware and recalibrate both pedals.

Check resistor-mode HIGH / pot-mode high impedance, ADC ranges, ladder taps, encoder, TRS, USB and BLE MIDI.

Each passive pedal input uses two resistors. Connect only unpowered variable-resistor or pot pedals.

This document does not establish ESD compliance, measured analog accuracy or simultaneous-supply qualification.

## Source documents checked

| | |
|---|---|
| **Arduino Nano ESP32 ABX00083 pinout** | Component-side top view, USB-C up; all 30 header pads |
| **Arduino Nano ESP32 schematic V0.3** | JP1: 1-15 top-down; JP2: 15-1 top-down |
| **Waveshare 24382 external P2 connector** | Local: design/fusion/datasheets/waveshare-24382-schematic.pdf |
| **Amphenol ACJS-MHDC socket drawing** | Local: design/fusion/datasheets/double-trs-ACJS-MHDC.pdf |
| **Pololu S13V10F5 / #4083** | 5V, 1A step-up/down regulator; retailer item 1178565 |
| **Pololu #2535** | 5mm addressable RGB LED / WS2811 / 5V / 800kHz |
| **Espressif self-powered USB devices** | VBUS sense: 100k / 150k divider to A4 / GPIO11 |
| **MIDI Association - MIDI 1.0 electrical specifications** | 5 V current-loop interface; TRS Type A pin mapping |

Source URLs, BOM, exact terminal connectivity and generated SVG sheets are in design/electrical/.